Swinburne University of Technology Hawthorn Campus
Department of Computing Technologies

COS20028 Big Data Architecture and Application
Lab Task 7 - *Semester 2, 2024*

Assessment Title: Retrieve Information with Hive.
Assessment Outcome: Pass/Fail
Assessable Item:

- One (1) piece of a report containing the complete HiveQL codes for retrieving the desired information with the screenshot of the execution results.

Purpose of Assignment

This assignment evaluates whether a student understands how to query the desired information via HiveQL in the HDFS environment.

This is an individual assignment and is mandatory for passing this unit.

Introduction

The source data is released on Canvas. You'll need to follow the instruction to connect your virtual machine with the secure FTP client software and then upload the source file onto the virtual machine before you can inject the data into HDFS for querying in HiveQL.
The source data "**austlang_dataset_nh.csv**" and "**austlang_dataset_nh.txt**" can be found in week 11 module on Canvas. You can follow the document entitled "Transferring files back and forth to the VM" in the COS20028 Tools module to setup the SFTP connection. The preparation steps for this lab task are listed as follows:

1. Check in your VM environment whether the folder "dataset" exists in the path "/home/training/training_materials/". If the folder doesn't exist, use the commands:
   a) ***cd ~/training_materials***
   b) ***mkdir dataset***
   to create the folder called "dataset".
2. Upload "austlang_dataset_nh.csv" and "austlang_dataset_nh.tsv" to the VM under the path "/home/training/training_materials/dataset" via SFTP.
3. Creating a database, a table, and import the csv file into MySQL by the following commands:
   a) ***mysql --user=training --password=training movielens***
   b) ***CREATE DATABASE indigenous;***
   c) ***CREATE TABLE austlang (lng_code VARCHAR(60 NOT NULL, lng_name VARCHAR(500),***

lng_synonym VARCHAR(2000), th_lng VARCHAR(2000), th_people VARCHAR(800), lng_lat VARCHAR(50), lng_lng VARCHAR(50), state_territory VARCHAR(20), uri VARCHAR(3000), PRIMARY KEY (lng_code));

d) ***LOAD DATA INFILE '/home/training/training_materials/dataset/austlang_dataset_nh.csv' INTO TABLE austlang FIELDS TERMINATED BY ',' ENCLOSED BY '"' LINES TERMINATED BY '\n';***

4. Creating a Hive database, a table, and import the csv file into Hive by the following commands:
   a) ***cd ~/training_materials/dataset***
   b) ***hdfs dfs -put austlang_dataset_nh.txt***
   c) ***hive***
   d) ***CREATE DATABASE indigenous;***
   e) ***CREATE TABLE IF NOT EXISTS indigenous.austlang (lng_code string, lng_name string, lng_synonym string, th_lng string, th_people string, th_lng_lat string, lng_lng string, state_territory string, uri string) ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' STORED AS TEXTFILE;***
   f) ***LOAD DATA INPATH 'austlang_dataset_nh.txt' INTO TABLE indigenous.austlang;***
   g) ***quit;***

The header of the input file with an example tuple is listed below:

Language_code	**Language_name**	**Language_synonym**	**Thesaurus_heading_language**	**Thesaurus_heading_people**	**Approximate_latitude_of_language_variety**	**Approximate_longitude_of_language_variety**	**State_territory**	**uri**
A1	Nyaki Nyaki / Njaki Njaki	Nyungar, Nyaki Nyaki, …	Nyaki Nyaki / Njaki Njaki language A1	Nyaki Nyaki / Njaki Njaki people A1	-32.3909	118.7551	WA	https://collection...

Now you have put the source data into both MySQL and Hive. You can start the tasks and answer the questions.

Assignment Goal

This assignment aims at training students to know how to import data into MySQL and the Hive in HDFS. An indigenous-related dataset is used in this assignment for students to understand better the general knowledge of the indigenous people in Australia. Students can also compare the differences between using an RDBMS and a simulated RDBMS environment on HDFS.

Assignment Task

For each task listed below, you are expected to implement queries for both MySQL and HiveQL. The following tasks should be accomplished:

1. Using MySQL and Hive queries to find out how many tuples are in the "austlang" table. Answer the question and provide the statement you used in the queries and the last page of the query result screenshot.
2. How many language names appear more than once in the lng_name attribute? List the language names with the appearance counts that appear more than once in all records. In

MySQL, rename the attributes to "language name" and "appearance".

3. Which sites own more than 230 data collected in this indigenous dataset? List the site and the count in your answer. A site can be a single state or a combination of multiple states. *Hint: if a tuple contains the state_territory value of "SA, VIC", it is considered as different from a single "SA" or a single "VIC".*
4. How many records match the criteria that the location is in the south of latitude -25.12 and the east of longitude 135.36?
5. How many times does "Mardudjara" appear in the lng_synonym attribute? List the statement and the answer obtained from both MySQL and HiveQL.
6. Does HiveQL return the result faster than MySQL? How many seconds do they take to return the results of Task 5?

Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Lab Task 7

Task	Assessment Outcome
Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional.	Pass/Fail

-- End of Assignment --